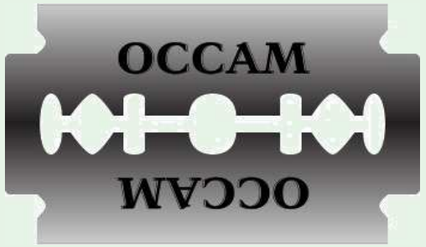
Review of Lecture 17

- Occam's Razor

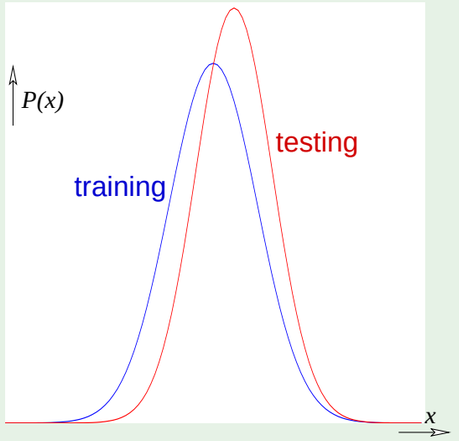
> The simplest model that fits the data is also the most plausible.



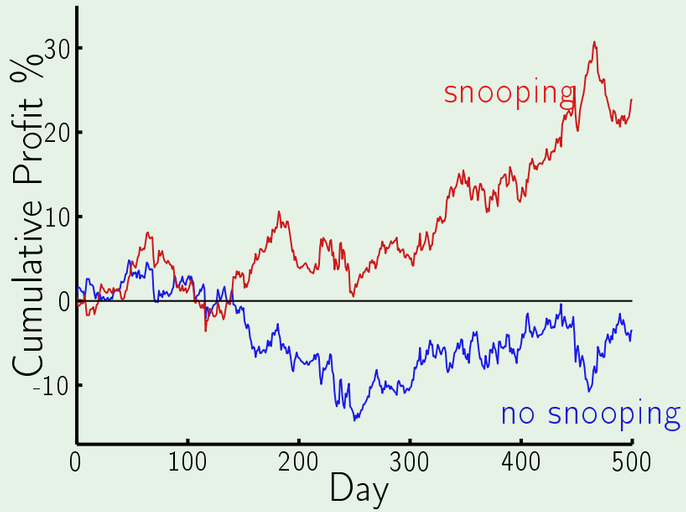
complexity of h $\longleftrightarrow$ complexity of $\mathcal{H}$

unlikely event $\longleftrightarrow$ significant if it happens

- Sampling bias



- Data snooping



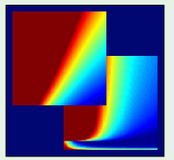
Learning From Data

Yaser S. Abu-Mostafa
California Institute of Technology

Lecture 18: Epilogue



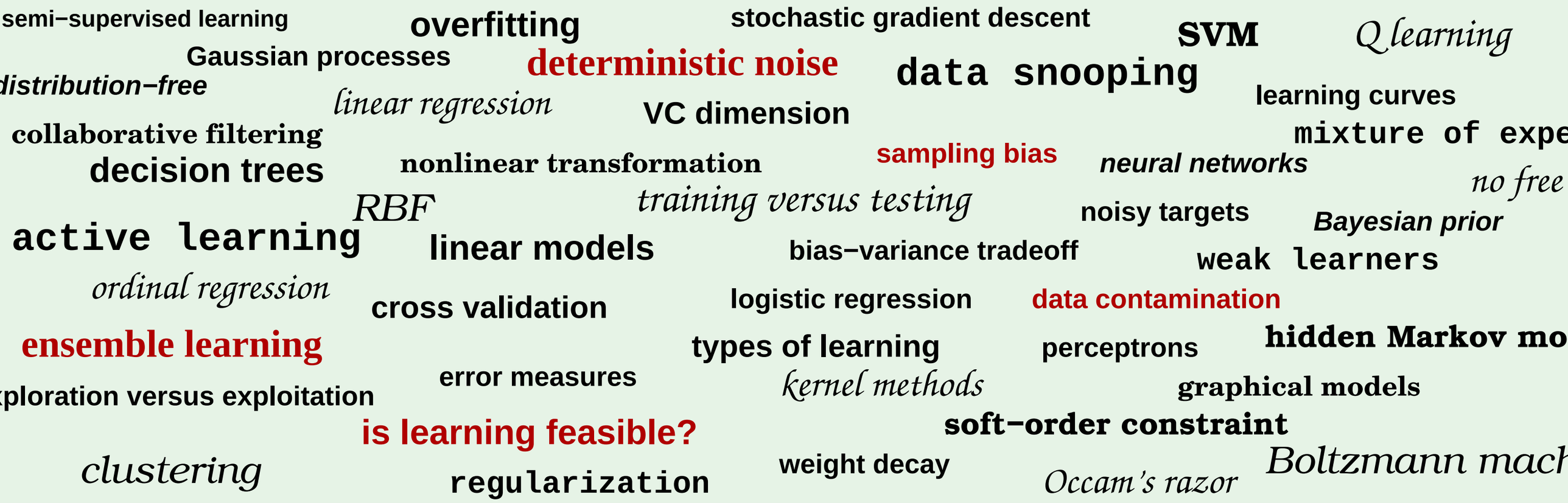
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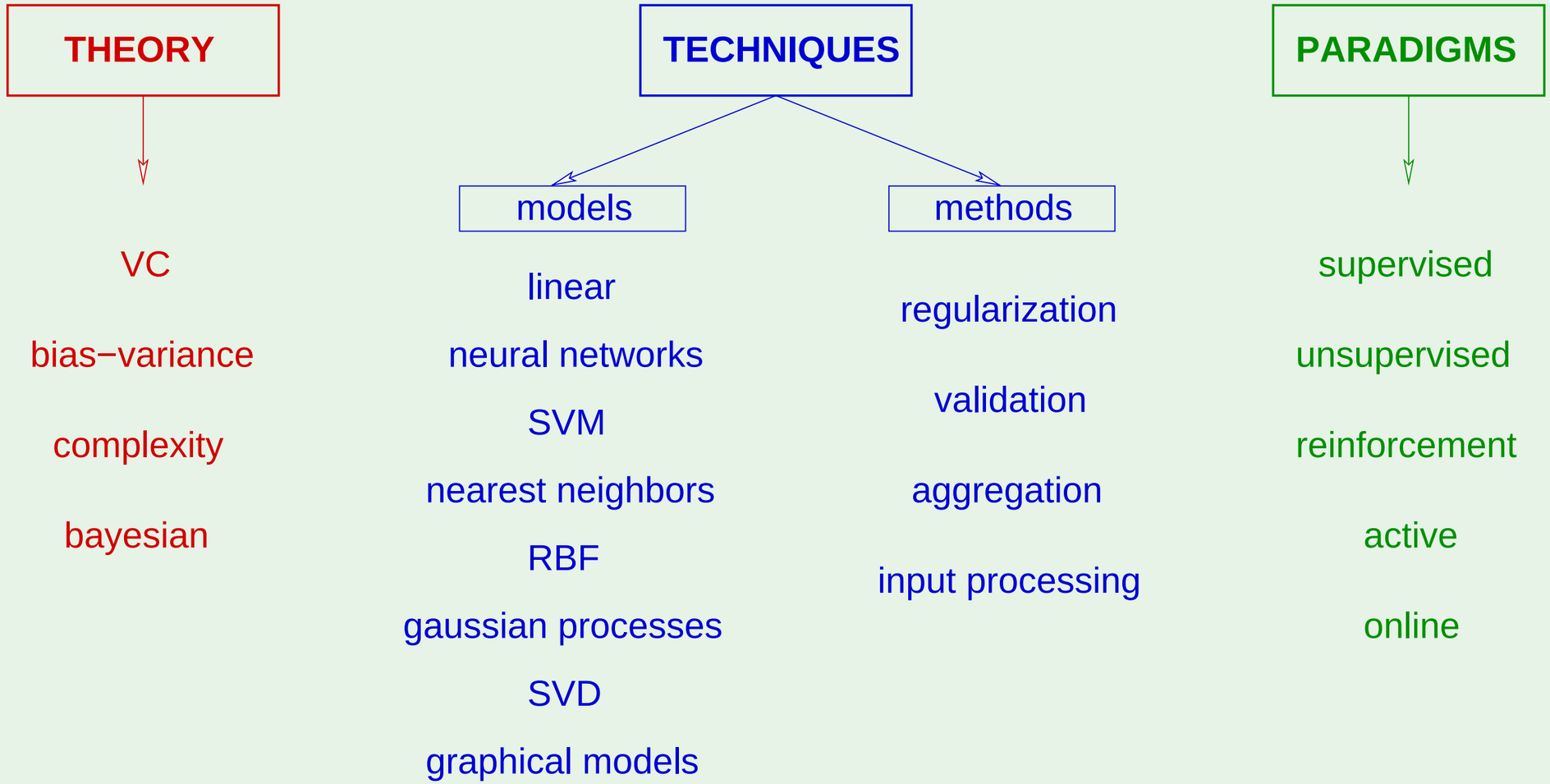
Outline

- The map of machine learning
- Bayesian learning
- Aggregation methods
- Acknowledgments

It's a jungle out there



The map


THEORY
VC
bias−variance
complexity
bayesian
TECHNIQUES
models
linear
neural networks
SVM
nearest neighbors
RBF
gaussian processes
SVD
graphical models
methods
regularization
validation
aggregation
input processing
PARADIGMS
supervised
unsupervised
reinforcement
active
online

Outline

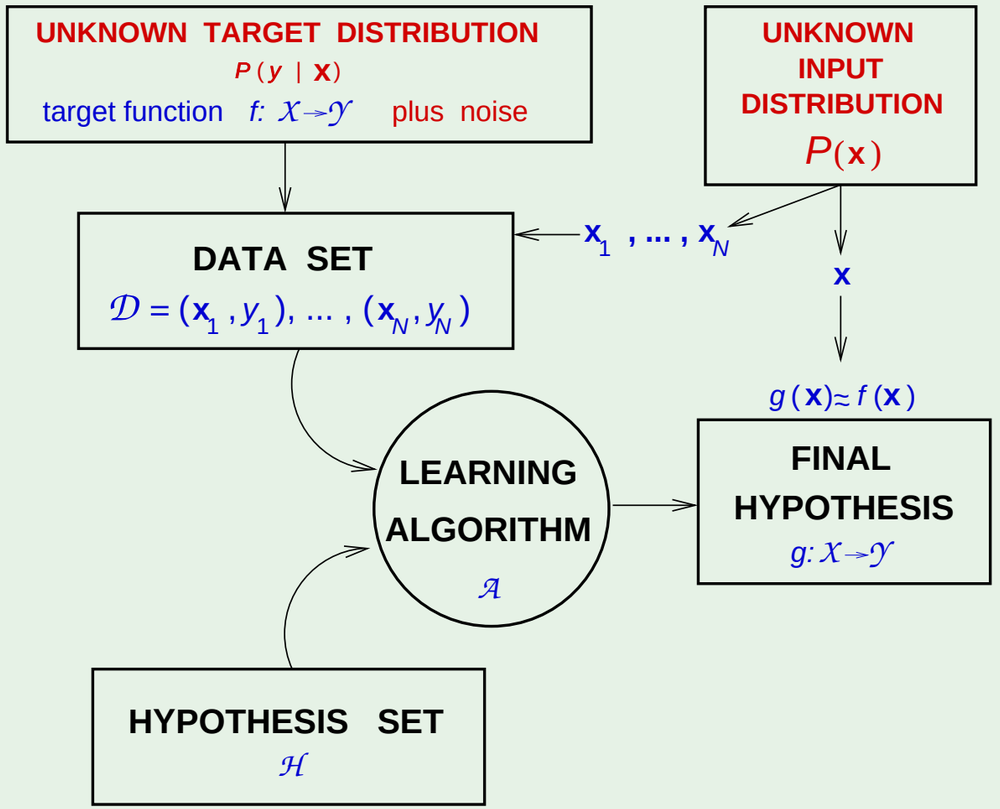
- The map of machine learning
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Probabilistic approach

Extend probabilistic role to all components

$P(\mathcal{D} \mid h = f)$ decides which h (likelihood)

How about $P(h = f \mid \mathcal{D})$?



The prior

$P(h = f \mid \mathcal{D})$ requires an additional probability distribution:

$$P(h = f \mid \mathcal{D}) \quad = \quad \frac{P(\mathcal{D} \mid h = f)\ P(h = f)}{P(\mathcal{D})} \quad \propto \quad P(\mathcal{D} \mid h = f)\ P(h = f)$$

$P(h = f)$ is the **prior**

$P(h = f \mid \mathcal{D})$ is the **posterior**

Given the prior, we have the full distribution

Example of a prior

Consider a perceptron: h is determined by $\mathbf{w} = w_0, w_1, \cdots, w_d$

A possible prior on $\mathbf{w}$: Each w_i is independent, uniform over $[-1, 1]$

This determines the prior over h - $P(h = f)$

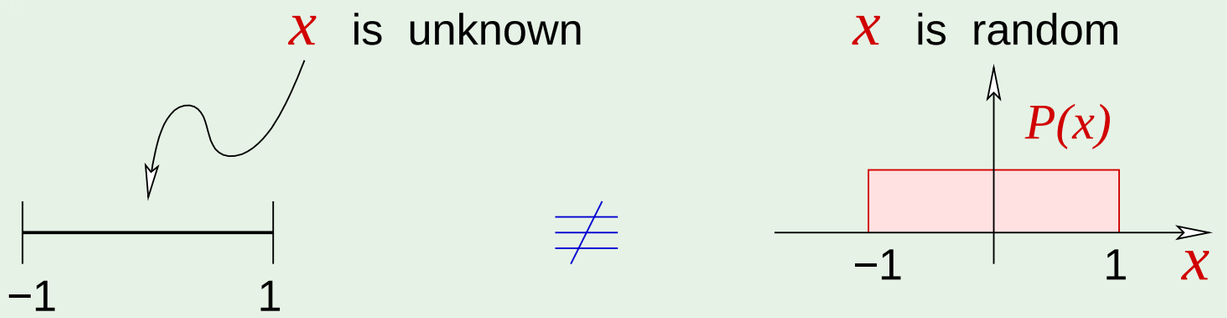
Given $\mathcal{D}$, we can compute $P(\mathcal{D} \mid h = f)$

Putting them together, we get $P(h = f \mid \mathcal{D})$

$$\propto \quad P(h = f) P(\mathcal{D} \mid h = f)$$

A prior is an assumption

Even the most "neutral" prior:



The true equivalent would be:



If we knew the prior

. . . we could compute $P(h = f \mid \mathcal{D})$ for every $h \in \mathcal{H}$

$\Longrightarrow$ we can find the most probable h given the data

we can derive $\mathbb{E}(h(\mathbf{x}))$ for every $\mathbf{x}$

we can derive the **error bar** for every $\mathbf{x}$

we can derive everything in a principled way

When is Bayesian learning justified?

1. The prior is **valid**

   trumps all other methods

2. The prior is **irrelevant**

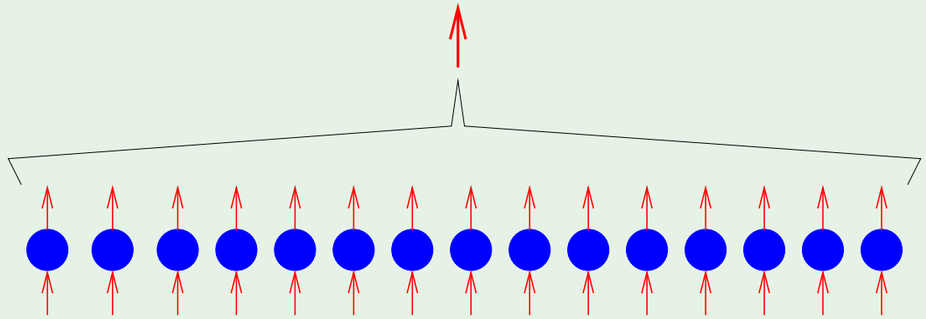
   just a computational catalyst

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What is aggregation?

Combining different solutions $h_1, h_2, \cdots, h_T$ that were trained on $\mathcal{D}$:



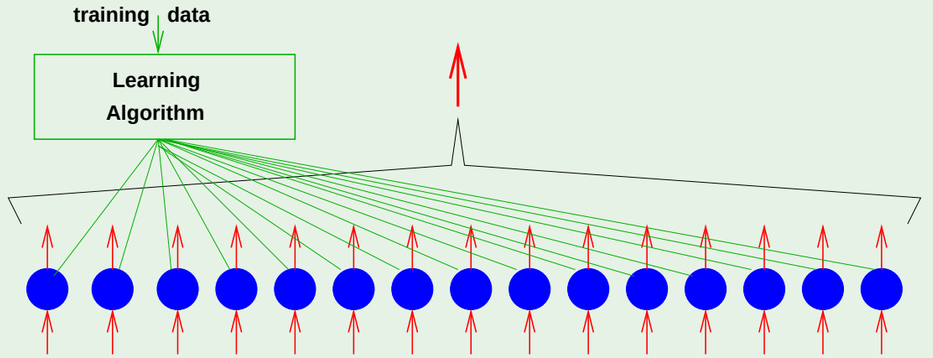
Regression: take an average

Classification: take a vote

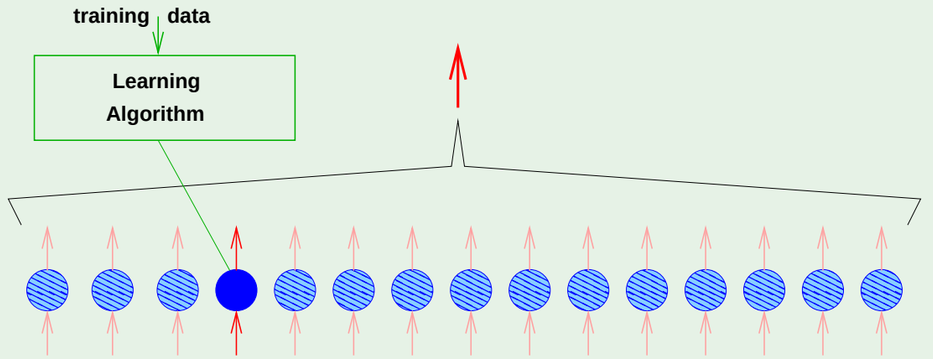
a.k.a. *ensemble learning* and *boosting*

Different from 2-layer learning

In a 2-layer model, all units learn **jointly**:



In aggregation, they learn **independently** then get combined:



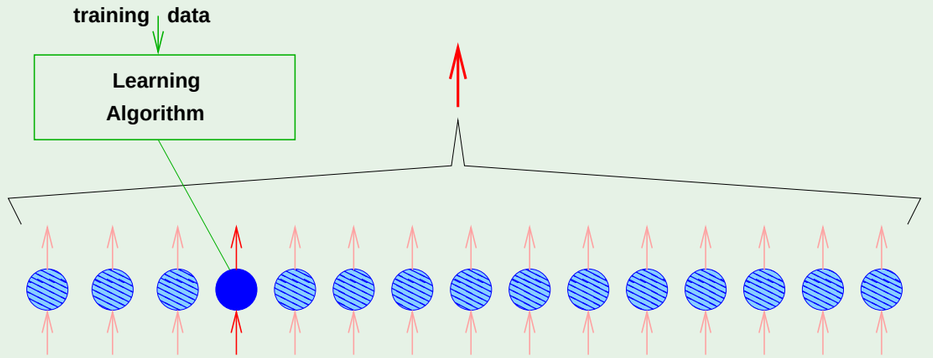
Two types of aggregation

1. **After the fact:** combines existing solutions

Example. Netflix teams merging "blending"

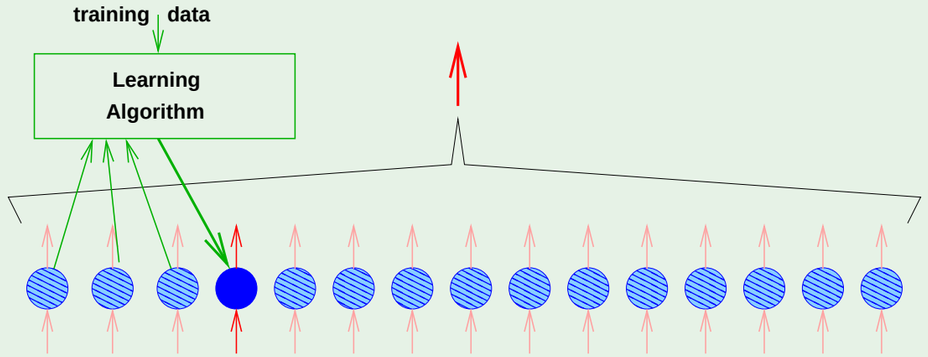
2. **Before the fact:** creates solutions to be combined

Example. Bagging - resampling $\mathcal{D}$



Decorrelation - boosting

Create $h_1, \cdots, h_t, \cdots$ sequentially: Make h_t decorrelated with previous h's:



Emphasize points in $\mathcal{D}$ that were misclassified

Choose weight of h_t based on $E_{\text{in}}(h_t)$

Blending - after the fact

For regression, $h_1, h_2, \cdots, h_T \longrightarrow g(\mathbf{x}) = \sum_{t=1}^{T} \alpha_t \, h_t(\mathbf{x})$

Principled choice of α_t's: minimize the error on an "aggregation data set" pseudo-inverse

Some α_t's can come out negative

Most valuable h_t in the blend?

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Course content

Professor **Malik Magdon-Ismail**, *RPI*

Professor **Hsuan-Tien Lin**, *NTU*

Course staff

Carlos Gonzalez (*Head TA*)

Ron Appel

Costis Sideris

Doris Xin

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To the fond memory of

Faiza A. Ibrahim